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Single-bundle patellar tendon versus non-anatomical double-bundle hamstrings ACL reconstruction: a prospective randomized study at 8-year minimum follow-up

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Abstract
Purpose The purpose of this study was to compare subjective, objective and radiographic outcome of the lateralized single-bundle bone-patellar tendon-bone autograft with a non-anatomical double-bundle hamstring tendons autograft anterior cruciate ligament (ACL) reconstruction technique at long-term follow-up.
Methods Seventy-nine non-consecutive randomized patients (42 men; 37 women) with unilateral ACL insufficiency were prospectively evaluated, before and after ACL reconstruction by means of the above-mentioned techniques, with a minimum follow-up of 8 years (range 8–10 years; mean 8.6 years). In the double-bundle hamstrings technique, we used one tibial and one femoral tunnel combined with one "over-the-top" passage, cortical staple's fixation and we left intact hamstrings' tibial insertion. Patients were evaluated subjectively and objectively, using IKDC score, Tegner level, manual maximum displacement test with KT-2000™ arthrometer. Radiographic evaluation was performed according to IKDC grading system, and re-intervention rate for meniscal lesions was also recorded.
Results The subjective and objective IKDC were similar in both groups while double-bundle hamstrings group showed significantly higher Tegner level ($P = 0.0007$), higher passive range of motion recovery ($P = 0.0014$), faster sport resumption ($P = 0.0052$), lower glide pivot-shift phenomenon ($P = 0.0302$) and lower re-intervention rate ($P = 0.0116$) compared with patellar tendon group. Radiographic evaluation showed significant lower objective degenerative changes in double-bundle hamstrings group at final follow-up ($P = 0.0056$).
Conclusion Although both techniques provide satisfactory results, double-bundle ACL reconstruction shows better functional results, with a faster return to sport activity, a lower re-operation rate and lower degenerative knee changes.

Keywords Knee · Arthroscopy · ACL reconstruction · Patellar tendon · Non-anatomical double-bundle hamstrings

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Introduction

Bone-patellar tendon-bone (BPTB) reconstruction is the most commonly used technique for anterior cruciate ligament (ACL) reconstruction and is should be considered as a basis of comparison for new procedures.

The recent rise of hamstring tendon grafts [5] is mainly based on the possibility of performing a double-bundle ACL reconstruction and to a lower post-operative morbidity compared with patellar tendon graft [25].

In a meta-analysis of 24 randomized controlled trials (RCTs) on single-bundle reconstruction, Biau et al. [2]

Table 1 Demographics and pre-operative psycho-activity evaluation

Demographics and pre-operative psycho-activity evaluation	Lateralized single-bundle bone-patellar tendon-bone	Non-anatomical double-bundle hamstrings
Number of patients = 79	39	40
Males = 42	20	22
Females = 37	19	18
Mean age	26 years (SD = 9.5)	27 years (SD = 9)
Mean height	176 cm (SD = 7.4)	175.4 cm (SD = 7.6)
Median height	175 cm (157-192 cm)	172 cm (163-195 cm)
Mean weight	78.8 kg (SD = 14.6)	78.4 kg (SD = 10.1)
Meniscus lesions	23	23
Chondral Outerbridge I°–II° lesions	1	1
From injury to surgery time	8.6 months (SD = 10.4)	8.9 months (SD = 10.4)
Pre-operative activity rating scale score (max 16)	13 (SD = 2.6)	13.2 (SD = 2.6)
Pre-operative psychovitality questionnaire score (max 18)	13 (SD = 3)	15 (SD = 3)

concluded that morbidity was lower for hamstring autografts than for patellar tendon autografts. In a recent study [1], the same authors reported that "postoperative knee instability was less common after ACL reconstruction with patellar tendon autograft than with hamstring tendon autograft".

Up to date, the only clinical study comparing the lateralized single-bundle BPTB autograft ACL reconstruction with a double-bundle hamstrings autograft was performed by Tsuda et al. [31]. The Authors concluded that the two techniques showed comparable results, but the study was not randomized and the final observation point was after a mean of 38 months from index surgery.

The purpose of this RCT study was to compare the subjective, objective, functional and radiographic results of the lateralized single-bundle BPTB autograft (LBPTB) with a non-anatomical double-bundle hamstring autograft (naDBH) technique at long-term follow-up (FU). The initial hypothesis was that naDBH group would have better results than LBPTB group 8 years minimum after surgery.

Materials and method

One hundred consecutive patients (50 men; 50 women) with unilateral ACL rupture were enrolled in the study. Inclusion criteria were a positive clinical examination (Lachman test, anterior drawer test and pivot-shift test) respect to contra-lateral normal knee. Patients with medial and/or lateral meniscal injuries, grade 1 or 2 medial collateral ligament (MCL) injuries and Outerbridge [23] grade 1 or 2 chondral lesions were also included.

Ten patients were excluded during surgery for partial posterior cruciate ligament (PCL) injury or Outerbridge grade 3–4 chondral lesions. Other three patients were excluded due to grade three MCL injuries.

Every patient was randomized to autologous lateralized single-bundle bone-patellar tendon-bone (LBPTB) or autologous non-anatomical double-bundle hamstrings (naDBH) ACL reconstruction technique using computer-generated randomization tables. All Patients were operated by the two senior authors or under their supervision between 1992 and 2000.

Approval was obtained from the Internal Review Board of Rizzoli Orthopaedic Institute according to the official guidelines of the Declaration of Helsinki 1996. All subjects were informed about the study procedure, the purpose of the study and any known risks; all of them provided their informed consent the day they were enrolled.

Of the 90 patients included, 79 were available at the 8-year minimum follow-up interval (range 8–10 years; mean 8.6 years). Forty patients were treated with the autologous naDBH procedure and thirty-nine patients underwent the autologous LBPTB technique.

The two groups were homogeneous regarding age, gender, weight and height, interval from injury to surgical treatment, sport activity level (as defined according to Activity Rating Scale [18]) and Psychovitality Questionnaire score [7]. Demographic and pre-operative psycho-activity data are summarized in Table 1.

Autologous lateralized single-bundle bone-patellar tendon-bone (lbptb) technique

BPTB autograft was harvested through a single straight midline incision. In all cases, we used central third of the ipsilateral patellar tendon. A 9-mm-large median portion of

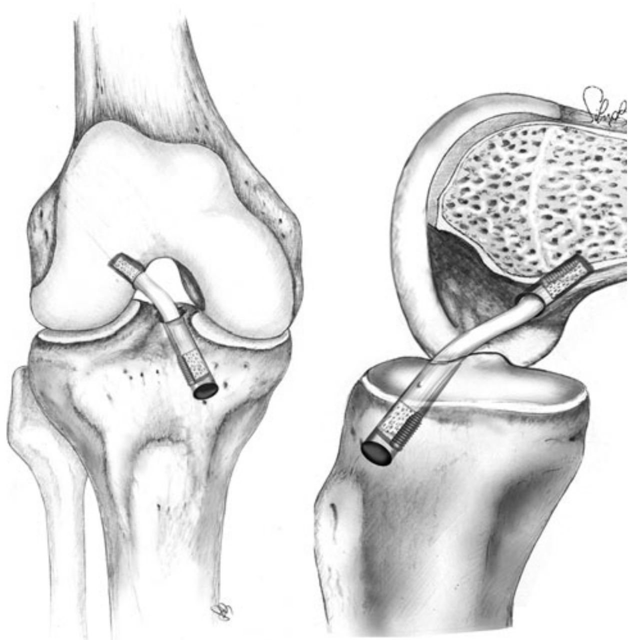

Fig. 1 Lateralized bone-patellar tendon-bone autograft ACL reconstruction (LBPTB)

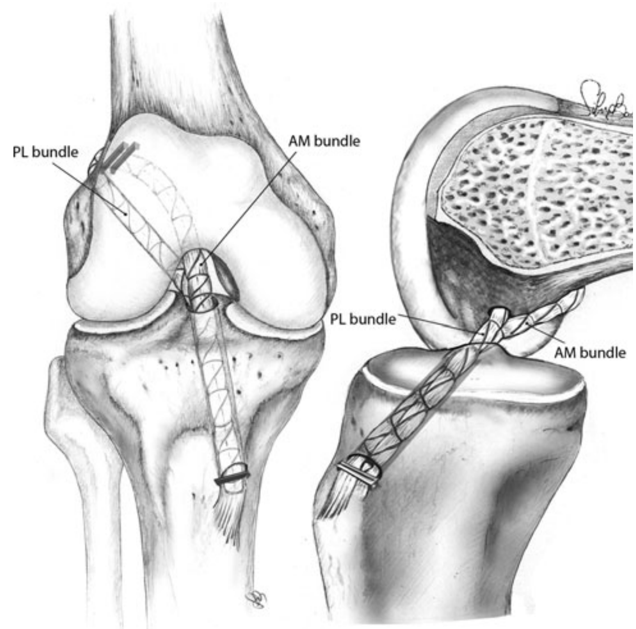

Fig. 2 Non-anatomical double-bundle hamstrings autograft ACL reconstruction (naDBH)

the tendon was removed together with proximal and distal bone plugs of 2.5 cm taken from patella and tibial tubercle, respectively.

A 9 mm tibial tunnel was drilled, according to Howell et al. [10] on tibial tunnel angle. The femoral tunnel was performed with the same size from the antero-medial portal at 10 o'clock position for the right knee and at 2 o'clock position for the left knee according to Loh and Fukuda [15].

Interference screws of 7–9 mm (Smith and Nephew Endoscopy, USA) were used to fix the graft in the femoral and tibial tunnel, respectively.

All grafts were pretensioned with 20 cycles and with maximum manual force applied fixed at 20° of knee flexion (Fig. 1).

Non-anatomical autologous double-bundle hamstrings (naDBH) technique

Semitendinosus and gracilis tendons (ST + G) from the ipsilateral limb were harvested with an open tendon stripper (Acufex, Microsurgical, Mansfield, MA, USA). The tibial insertion of the two tendons was left intact. The harvested tendons were then tightly sutured at their free proximal ends to obtain a sufficient strength for traction and to allow easy passage of the tendons through the tibial and the femoral drill holes.

The tibial tunnel was performed starting close to the MCL with an exit point on the postero-medial aspect of the native ACL footprint.

The femoral tunnel was performed from the medial portal with the knee flexed to approximately 120°. The entry point of the tunnel was always below the "resident ridge", about 10 mm anteriorly to the posterior margin, close to anatomical postero-lateral bundle insertion. The exit point in the lateral aspect of the femur was immediately above the end of the lateral femoral epicondyle, at about 5 mm from the "over-the-top" position.

After a lateral incision, the tendons were passed "over-the-top", than from outside in the femoral tunnel and back again in the tibial tunnel to obtain a double-bundle reconstruction.

The graft was tensioned and the knee cycled through a full range motion approximately 20 times.

The "over-the-top" antero-medial bundle was fixed at 90° and secured with 2 staples at the lateral condyle, while the postero-lateral bundle was fixed at 15° and secured with 1 staple at the anterior tibial cortex or otherwise with a transosseus suture knot, according to the length of the autologous tendons.

This technique attempts to reproduce the kinematic effects of both antero-medial and postero-lateral bundle of the ACL with a four-bundle reconstruction [16] (Fig. 2).

There were no significant statistical differences in the incidence of associated meniscus and chondral injuries between the two groups (n.s.) (Table 2).

Table 2 ACL-associated lesions and their treatment at the time of the surgery

ACL-associated lesions/treatments	Lateralized single-bundle bone-patellar tendon-bone	Non-anatomical double-bundle hamstrings
Medial meniscus tear	12	14
Lateral meniscus tear	8	7
Medial + lateral menisci tear	3	2
Selective meniscectomy	23	22
Meniscal suture	0	1
Grade I–II MCL tear (no reparation performed)	4	4

All patients of our series underwent the same post-operative rehabilitation protocol, regardless of the presence of associated injuries. No brace was used. Range of motion (ROM), quadriceps muscle active exercises and straight leg raises were started on the first post-operative day with isometric quadriceps contractions and progressed to active closed chain exercise. Functional muscle stimulation was used 2 h three times a day for the first 4 weeks after surgery. Patients were allowed to partial weight bearing with no braces during the first 2 weeks. Full passive extension and active flexion over a range of 0°–120° was started from the third post-operative day in both isometric and isotonic fashion. Full weight bearing was allowed from the third week. Stationary biking, active knee extensions with weights applied, one quarter squatting and proprioceptive exercises were introduced at 4 weeks after intervention. After 1 month, isotonic and closed chain exercises were started. All exercises were done under continuous direction of a physical therapist, to control the individual compliance to the standard protocol. Running was recommended after 2 months, and cutting and lateral sports were allowed 3–4 months after surgery depending on their performance level. The criteria to allow sport resumption were isokinetic tests with less than 10% difference between healthy and operated knee, muscle atrophy of operated leg equal or less than 1 cm inferior compared to contra-lateral leg, one leg hop more than 90% and good firm anterior tibial stop at objective clinical evaluation. The decision regarding sport resumption timing was always taken in combination by the surgeon, physical therapist and patient.

Patient evaluation

All patients were clinically and instrumentally evaluated pre-operatively and at a minimum FU of 8 years (range 8–10 years; mean 8.6 years) by four independent observers (2 foreign fellows and 2 residents) without any other involvement in our study. To obtain a single-blinded evaluation of our series, we asked to an independent nurse to completely cover scars on the harvesting and fixation sites with doilies disposed in the same way, regardless of the surgical technique.

The subjective IKDC evaluation questionnaire (International Knee Documentation Committee 2000) [11], not yet released at the begin of the study, was administered at final FU control. For clinical objective evaluation, the objective IKDC knee examination form (International Knee Documentation Committee 1991) [9] was prospectively recorded. The functional capability of the knee before injury, prior to surgery and at the 8-year minimum observation time point was evaluated using the Tegner Score [30].

A KT-2000^{TM} arthrometer (MEDmetric, San Diego, CA, USA) was used for laxity measurements.

Pre-operatively a bilateral long-standing antero-posterior (AP) weight-bearing roentgenogram and a lateral view in 30° of flexion were performed to evaluate patient alignment. A bilateral double-leg postero-anterior (PA) weight-bearing roentgenogram at 35° to 45° of flexion (tunnel view) and a lateral view in 30° of flexion were taken at FU controls. Radiographs were evaluated and graded by a single expert radiologist, blinded to patient identification and surgical status, using the IKDC Radiographic Grading System [19].

Time to return to sport, anterior knee pain, re-intervention for meniscal lesions and/or revision surgery during the FU period were also recorded.

Statistical analysis

Differences between the two groups for subjective IKDC, KT2000^{TM} evaluation and sport resumption were evaluated with unpaired Student's t test. For differences in Tegner level, we used the non-parametric Mann–Whitney test. For differences in objective IKDC form and radiographic IKDC grade, manual pivot-shift test, incidence of re-intervention for meniscal lesions and anterior knee pain the chi-square test was used. The level of significance was set at $P < 0.05$. Post hoc power analysis on the performed statistics showed a value >0.8 for all demographic and

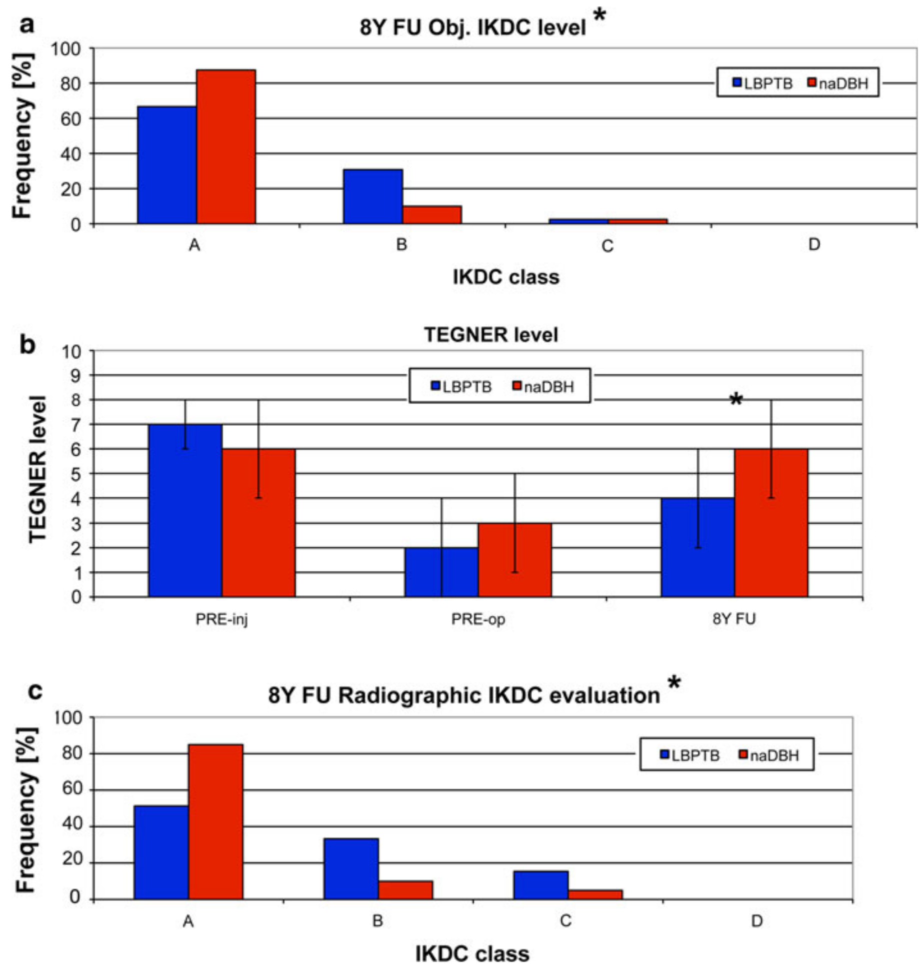

Fig. 3 **a** objective IKDC level at minimum 8 years after surgery; **b** Tegner level prospective evaluation (median values ± interquartile range); **c** IKDC radiographic evaluation for degenerative joint changes at minimum 8 years after surgery. Statistical significant differences between our two groups ($P < 0.05$) are marked with *asterisks* (see "Results" for details)

Table 3 Manual pivot-shift test at minimum 8 years after surgery

Manual pivot-shift test	Lateralized single-bundle bone-patellar tendon-bone	Non-anatomical double-bundle hamstrings	P value
Negative	26 (66%)	36 (90%)	0.0302
Glide (+)	12 (31%)	3 (7%)	
Moderately positive (++)	1 (3)%	1 (3%)	
Severely positive (+++)	0 (0%)	0 (0%)	

psycho-activity data tests. Statistical analysis was performed using Analyse-it-2.00 (Analyse-it Software, Ltd, Leeds, UK).

Reported results are expressed in terms of mean value ± standard deviation for continuous data and median ± interquartile range for non-continuous data.

Results

No statistically significant differences were detected between the two groups regarding mean subjective IKDC evaluation at final FU (LBPTB group 82 ± 20; naDBH group 88 ± 9; n.s.).

The objective IKDC evaluation revealed a higher percentage of normal knees in the naDBH group compared with the LBPTB group at final FU, but this trend was not statistically significant ($P = 0.0701$) (Fig. 3a).

The pivot-shift test was significantly better in the naDBH group than in the LBPTB group ($P = 0.0302$) (Table 3).

Mean passive ROM ($P = 0.0014$) was significantly higher for naDBH group (146.5 ± 5.5°) than for LBPTB group (140.9 ± 5.8°) at final follow-up evaluation.

Table 4 Anterior knee pain, mean time for sport resumption and incidence of re-intervention for meniscal lesions at minimum 8 years after surgery

	Lateralized single-bundle bone-patellar tendon-bone		Non-anatomical double-bundle hamstrings		P value
Anterior knee pain	No	Yes	No	Yes	n.s.
	25 (64%)	14 (36%)	32 (80%)	8 (20%)	
Mean time for sport resumption	6.6 months (SD = 5.2)		3.8 months (SD = 1.9)		0.0052
Incidence of re-intervention for meniscal lesions	13 (33%)		4 (10%)		0.0116

The Tegner level at final FU was significantly higher for naDBH group than for LBPTB group ($P = 0.0007$), starting from comparable pre-injury (n.s.) and pre-operative (n.s.) situations (Fig. 3b).

The side-to-side difference in manual maximum displacement test performed with a KT-2000™ arthrometer revealed no significant differences between the two groups at 8 years of minimum FU (LBPTB group 0.4 ± 0.6 mm; naDBH group 1.1 ± 1.9 mm; n.s.).

Post-operative radiographic evaluation revealed lower statistically significant degenerative joint changes in naDBH group compared to LBPTB group after 8 years of minimum follow-up, according to IKDC grading system (Fig. 3c).

No statistically significant differences were found between the two groups regarding anterior knee pain (n.s.). Patients of naDBH group returned to sport activities significantly faster compared to those of LBPTB group ($P = 0.0052$), and we found a significant higher value for re-intervention rate due to newly occurred meniscal lesions in LBPTB group compared to naDBH group ($P = 0.0116$). These data were summarized in Table 4.

In both groups, we had no graft failure and no complications or adverse events related to surgical procedure.

Discussion

The main result of this study is that naDBH provides superior outcomes regarding pivot-shift restraint and degenerative joint changes prevention compared to LBPTB at a minimum follow-up of 8 years.

Many in vitro and clinical studies suggest that double-bundle ACL reconstruction can improve post-operative rotational stability compared to single-bundle techniques [12, 21, 28, 34].

In vivo biomechanical studies have reported that a laterally placed femoral tunnel (from the 11 o'clock position to the 10 o'clock position for a right knee) in single-bundle ACL reconstruction provides a kinematic restoration comparable to a double-bundle graft [13, 35].

Tsuda and colleagues [31] have found comparable clinical results between lateralized single-bundle bone-patellar tendon-bone (LBPTB) and anatomical double-bundle hamstrings (DBH) ACL reconstructions. The Authors anyway compared these two techniques at short-term FU, in a non-randomized study, without a radiographic assessment for degenerative joint changes and they did not analyse the re-intervention rate due to meniscal lesions and the degenerative joint changes at the FU period.

The described naDBH technique is easier and less invasive than more anatomical double-bundle ACL reconstructions where two femoral and tibial tunnels are performed. In our prospective RCT long-term FU study, naDBH group was significantly superior compared to LBPTB group in Tegner score, mean time to sport resumption and manual pivot-shift test evaluation at 8 years minimum FU. However, subjective/objective IKDC scores and anterior knee laxity KT-2000 measurements were similar in both groups. Our IKDC, pivot-shift and anterior laxity results are similar to those reported by Järvelä [12] comparing single-bundle and anatomical double-bundle hamstrings autografts.

Radiographic osteoarthritis and re-intervention rate due to post-operative meniscal lesions were significantly higher in LBPTB group. Our IKDC grading system results are similar to those published by Pinczewski et al. [25] regarding degenerative radiographic changes at long-term FU in BPTB vs. 4-strand hamstrings autografts. The re-intervention rate for meniscal injury is not largely investigated in literature for double-bundle technique, and results for BPTB ACL reconstruction are variable [6, 32]. The low incidence of meniscal injury in our naDBH group could be related to the better dynamic stability provided by our technique.

Mean passive ROM at final follow-up evaluation was significantly higher for our naDBH group than for LBPTB group. Even though the clinical impact of such few degrees may be doubtful, Shelbourne has also shown how ROM full restoration is one of the key factors in ACL reconstruction success [27].

The conclusions of this work must take into account the several limitations of the study.

This paper evaluates two ACL reconstruction techniques (LBPTB vs. non-anatomical naDBH) differing for graft type and fixation devices. This variables might have contributed to some of the differences in the results.

Unlike previous published meta-analyses [2, 8, 26], a recent work by Biau et al. [1] has shown similar or superior clinical results for patellar tendon compared with hamstrings.

The graft tensioning might contribute to the ligamentization process of the implant and to the long-term clinical and radiological results [17]. Both grafts were manually tensioned. The ideal combination of knee angles and graft tension at the fixation is standardized for BPTB reconstruction, but has not been clearly established for double-bundle techniques [4, 33].

The described original technique for double-bundle ACL reconstruction (naDBH) is non-anatomical, even if more simple, reproducible, time and cost saving. Petersen et al. [24] in 2007 have performed an in vitro study concluding that a double-bundle ACL reconstruction with two tibial tunnels provides a better laxity control compared to a single tibial tunnel technique. Moreover, Zantop et al. [37] have evaluated the double-bundle technique with different postero-lateral femoral tunnel positions, concluding that an anatomical tunnel placement of the antero-medial and postero-lateral bundles is superior to a non-anatomical femoral tunnel position. Nevertheless, Zaffagnini et al. [36] in an in vitro study have shown that femoral tunnel orientation is an important variable also in a non-anatomical technique. We finally agree with Tashman et al. [29], who concluded that “it has been difficult to predict clinical outcome based on the results of cadaver studies, which cannot replicate functional loading and represent only the time-zero condition (immediately after graft fixation)” in a laboratory setting.

A quantitative evaluation regarding the static anterior knee laxity was performed using the KT-2000™ arthrometer. Dynamic rotational stability was assessed only by the manual pivot-shift test. At the state of the art, this is the better way to test dynamic knee stability [3, 22], and it is significantly correlated with subjective satisfaction, overall knee function and sports participation [14]. On the other hand, this test is very subjective and observer dependent, and it is not correctly numerically quantifiable.

Even if a recent meta-analysis study was proposed by Meredick et al. [20] in order to definitively define which ACL reconstruction technique can provide the better outcomes, it remains extremely hard to obtain a true answer due to the different evidence levels of the presented studies and to the large amount of variables that must be considered in such a study.

Conclusion

This study with the standard actual available evaluation methods was able to show similar results for subjective and objective IKDC, but a significantly better Tegner activity level, better range of motion, reduced glide pivot-shift phenomenon and degenerative changes and reoperation rate for our non-anatomical double-bundle hamstring autograft compared with a lateralized single-bundle BPTB autograft ACL reconstruction at minimum follow-up of 8 years.

Based on available results, the possibility to improve function while decreasing the risk of degenerative changes and reoperations related to ACL reconstruction led us to believe that in case of ACL lesions the use of a double-bundle hamstrings reconstruction would help to maintain patient activity level for a longer time while protecting the joint against degeneration. This study supports that belief; hence, we recommend the use of the described technique, especially in athletes.

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Conflict of interest The authors declare that they have no conflict of interest.

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